## Привіт, мене зовуть Владислав

# І я починаю писати код в Python

Ця програма робить щось корисне

# Вступ 1.1

In [3]:
```
print ("Hello World!")
```

```
Hello World!
```

In [4]:
```
    120 + 20
```

Out[4]: 140

## 1.1 Розділ

In [12]:
```
ls
```

```
 Volume in drive C has no label.
 Volume Serial Number is 30EC-41E0

 Directory of C:\Users\vshpa

21.10.2026  15:19    <DIR>          .
06.02.2026  18:31    <DIR>          ..
19.10.2026  16:12    <DIR>          .anaconda
19.10.2026  16:05    <DIR>          .anaconda-desktop
22.05.2026  20:30    <DIR>          .android
19.10.2026  16:22    <DIR>          .conda
19.10.2026  16:11                39 .condarc
21.10.2026  15:07    <DIR>          .ipynb_checkpoints
19.10.2026  17:42    <DIR>          .ipython
19.10.2026  16:49    <DIR>          .jupyter
19.10.2026  16:05    <DIR>          .mamba
19.10.2026  17:18               145 .python_history
03.01.2026  07:26    <DIR>          ansel
03.01.2026  06:22    <DIR>          Contacts
19.10.2026  18:12    <DIR>          Documents
21.10.2026  15:09    <DIR>          Downloads
03.01.2026  06:22    <DIR>          Favorites
21.10.2026  15:19             3я791 first.ipynb
03.01.2026  07:51    <DIR>          iCloudDrive
03.01.2026  06:57    <DIR>          iCloudPhotos
03.01.2026  06:22    <DIR>          Links
19.10.2026  16:39    <DIR>          miniforge3
21.04.2026  22:56    <DIR>          Music
16.02.2026  20:38    <DIR>          OneDrive
19.10.2026  17:44    <DIR>          PycharmProjects
25.04.2026  03:57    <DIR>          Saved Games
03.01.2026  06:35    <DIR>          Searches
21.10.2026  15:07               689 Untitled.ipynb
21.10.2026  14:53                17 untitled.py
21.10.2026  14:55                 0 untitled1.py
19.10.2026  17:57    <DIR>          UntitledFolder
02.10.2026  00:32    <DIR>          Videos
               6 File(s)          4я681 bytes
              26 Dir(s)  81я730я244я608 bytes free
```

In [33]:
```
import time

for l in range (10):
    print (l)
    time.sleep(3)
```

```
0
1
2
3
4
5
6
7
8
9
```

In [25]:
```
my_list = [1, 2, 3, 4]
```

In [27]:
```
my_list
```

Out[27]: [1, 2, 3, 4]

In [28]:
```
[x**2 for x in range (-10, 11)]
```

Out[28]: [100, 81, 64, 49, 36, 25, 16, 9, 4, 1, 0, 1, 4, 9, 16, 25, 36, 49, 64, 81, 100]

In [30]:
```
s = "This game is briliant"
```

In [31]:
```
print (s)
```

```
This game is briliant
```

In [37]:
```
pip install jupyterlab_execute_time
```

```
Requirement already satisfied: jupyterlab_execute_time in .\AppData\Local\Programs
\Python\Python313\Lib\site-packages (3.3.0)
Requirement already satisfied: jupyterlab<5,>=4 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from jupyterlab_execute_time) (4.6.4)
Requirement already satisfied: async-lru>=1.0.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_time) (2.
4.0)
Requirement already satisfied: httpx<1,>=0.25.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_time) (0.2
8.1)
Requirement already satisfied: ipykernel!=6.30.0,>=6.5.0 in .\AppData\Local\Progra
ms\Python\Python313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_t
ime) (7.4.0)
Requirement already satisfied: jinja2>=3.0.3 in .\AppData\Local\Programs\Python\Py
thon313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_time) (3.1.6)
Requirement already satisfied: jupyter-builder>=1.0.2 in .\AppData\Local\Programs
\Python\Python313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_tim
e) (1.2.3)
Requirement already satisfied: jupyter-core in .\AppData\Local\Programs\Python\Pyt
hon313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_time) (5.9.1)
Requirement already satisfied: jupyter-lsp>=2.0.0 in .\AppData\Local\Programs\Pyth
on\Python313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_time)
(2.3.1)
Requirement already satisfied: jupyter-server<3,>=2.19.0 in .\AppData\Local\Progra
ms\Python\Python313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_t
ime) (2.21.1)
Requirement already satisfied: jupyterlab-server<3,>=2.28.0 in .\AppData\Local\Pro
grams\Python\Python313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execut
e_time) (2.28.1)
Requirement already satisfied: notebook-shim>=0.2 in .\AppData\Local\Programs\Pyth
on\Python313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_time)
(0.2.4)
Requirement already satisfied: packaging>=23.2 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_time) (26.
3)
Requirement already satisfied: tornado>=6.2.0 in .\AppData\Local\Programs\Python\P
ython313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_time) (6.5.1
0)
Requirement already satisfied: traitlets in .\AppData\Local\Programs\Python\Python
313\Lib\site-packages (from jupyterlab<5,>=4->jupyterlab_execute_time) (5.16.1)
Requirement already satisfied: anyio in .\AppData\Local\Programs\Python\Python313
\Lib\site-packages (from httpx<1,>=0.25.0->jupyterlab<5,>=4->jupyterlab_execute_ti
me) (4.15.1)
Requirement already satisfied: certifi in .\AppData\Local\Programs\Python\Python31
3\Lib\site-packages (from httpx<1,>=0.25.0->jupyterlab<5,>=4->jupyterlab_execute_t
ime) (2026.10.18)
Requirement already satisfied: httpcore==1.* in .\AppData\Local\Programs\Python\Py
thon313\Lib\site-packages (from httpx<1,>=0.25.0->jupyterlab<5,>=4->jupyterlab_exe
cute_time) (1.0.9)
Requirement already satisfied: idna in .\AppData\Local\Programs\Python\Python313\L
ib\site-packages (from httpx<1,>=0.25.0->jupyterlab<5,>=4->jupyterlab_execute_tim
e) (3.20)
Requirement already satisfied: h11>=0.16 in .\AppData\Local\Programs\Python\Python
313\Lib\site-packages (from httpcore==1.*->httpx<1,>=0.25.0->jupyterlab<5,>=4->jup
yterlab_execute_time) (0.16.0)
Requirement already satisfied: argon2-cffi>=21.1 in .\AppData\Local\Programs\Pytho
n\Python313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->j
upyterlab_execute_time) (25.1.0)
Requirement already satisfied: jupyter-client>=7.4.4 in .\AppData\Local\Programs\P
ython\Python313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupyterlab<5,>=
4->jupyterlab_execute_time) (8.10.0)
```

```
Requirement already satisfied: jupyter-events>=0.11.0 in .\AppData\Local\Programs
\Python\Python313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupyterlab<5,
>=4->jupyterlab_execute_time) (0.12.1)
Requirement already satisfied: jupyter-server-terminals>=0.4.4 in .\AppData\Local
\Programs\Python\Python313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupy
terlab<5,>=4->jupyterlab_execute_time) (0.5.4)
Requirement already satisfied: nbconvert>=6.4.4 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->ju
pyterlab_execute_time) (7.17.2)
Requirement already satisfied: nbformat>=5.3.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->ju
pyterlab_execute_time) (5.11.1)
Requirement already satisfied: prometheus-client>=0.9 in .\AppData\Local\Programs
\Python\Python313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupyterlab<5,
>=4->jupyterlab_execute_time) (0.26.0)
Requirement already satisfied: pywinpty!=3.0.4,>=2.0.1 in .\AppData\Local\Programs
\Python\Python313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupyterlab<5,
>=4->jupyterlab_execute_time) (3.1.0)
Requirement already satisfied: pyzmq>=24 in .\AppData\Local\Programs\Python\Python
313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyterla
b_execute_time) (27.2.0)
Requirement already satisfied: send2trash>=1.8.2 in .\AppData\Local\Programs\Pytho
n\Python313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->j
upyterlab_execute_time) (2.1.0)
Requirement already satisfied: terminado>=0.8.3 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->ju
pyterlab_execute_time) (0.18.1)
Requirement already satisfied: websocket-client>=1.7 in .\AppData\Local\Programs\P
ython\Python313\Lib\site-packages (from jupyter-server<3,>=2.19.0->jupyterlab<5,>=
4->jupyterlab_execute_time) (1.9.2)
Requirement already satisfied: babel>=2.10 in .\AppData\Local\Programs\Python\Pyth
on313\Lib\site-packages (from jupyterlab-server<3,>=2.28.0->jupyterlab<5,>=4->jupy
terlab_execute_time) (2.18.0)
Requirement already satisfied: json5>=0.9.0 in .\AppData\Local\Programs\Python\Pyt
hon313\Lib\site-packages (from jupyterlab-server<3,>=2.28.0->jupyterlab<5,>=4->jup
yterlab_execute_time) (0.17.3)
Requirement already satisfied: jsonschema>=4.18.0 in .\AppData\Local\Programs\Pyth
on\Python313\Lib\site-packages (from jupyterlab-server<3,>=2.28.0->jupyterlab<5,>=
4->jupyterlab_execute_time) (4.26.0)
Requirement already satisfied: requests>=2.31 in .\AppData\Local\Programs\Python\P
ython313\Lib\site-packages (from jupyterlab-server<3,>=2.28.0->jupyterlab<5,>=4->j
upyterlab_execute_time) (2.34.2)
Requirement already satisfied: typing_extensions>=4.16.0 in .\AppData\Local\Progra
ms\Python\Python313\Lib\site-packages (from anyio->httpx<1,>=0.25.0->jupyterlab<5,
>=4->jupyterlab_execute_time) (4.16.0)
Requirement already satisfied: argon2-cffi-bindings in .\AppData\Local\Programs\Py
thon\Python313\Lib\site-packages (from argon2-cffi>=21.1->jupyter-server<3,>=2.19.
0->jupyterlab<5,>=4->jupyterlab_execute_time) (26.1.0)
Requirement already satisfied: comm>=0.1.1 in .\AppData\Local\Programs\Python\Pyth
on313\Lib\site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<5,>=4->jupyter
lab_execute_time) (0.2.3)
Requirement already satisfied: debugpy>=1.6.5 in .\AppData\Local\Programs\Python\P
ython313\Lib\site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<5,>=4->jupy
terlab_execute_time) (1.8.22)
Requirement already satisfied: ipython>=7.23.1 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<5,>=4->ju
pyterlab_execute_time) (9.17.1)
Requirement already satisfied: matplotlib-inline>=0.1 in .\AppData\Local\Programs
\Python\Python313\Lib\site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<5,
>=4->jupyterlab_execute_time) (0.2.2)
Requirement already satisfied: nest-asyncio2>=1.7.0 in .\AppData\Local\Programs\Py
```

```
thon\Python313\Lib\site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<5,>=4
->jupyterlab_execute_time) (1.7.4)
Requirement already satisfied: colorama>=0.4.4 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jup
yterlab<5,>=4->jupyterlab_execute_time) (0.4.6)
Requirement already satisfied: ipython-pygments-lexers>=1.0.0 in .\AppData\Local\P
rograms\Python\Python313\Lib\site-packages (from ipython>=7.23.1->ipykernel!=6.30.
0,>=6.5.0->jupyterlab<5,>=4->jupyterlab_execute_time) (1.1.1)
Requirement already satisfied: jedi>=0.18.2 in .\AppData\Local\Programs\Python\Pyt
hon313\Lib\site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyter
lab<5,>=4->jupyterlab_execute_time) (0.20.1)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in .\AppData\Local\Pr
ograms\Python\Python313\Lib\site-packages (from ipython>=7.23.1->ipykernel!=6.30.
0,>=6.5.0->jupyterlab<5,>=4->jupyterlab_execute_time) (3.0.53)
Requirement already satisfied: psutil>=7 in .\AppData\Local\Programs\Python\Python
313\Lib\site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab
<5,>=4->jupyterlab_execute_time) (7.2.2)
Requirement already satisfied: pygments>=2.14.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jup
yterlab<5,>=4->jupyterlab_execute_time) (2.21.0)
Requirement already satisfied: stack_data>=0.6.0 in .\AppData\Local\Programs\Pytho
n\Python313\Lib\site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->ju
pyterlab<5,>=4->jupyterlab_execute_time) (0.6.3)
Requirement already satisfied: wcwidth>=0.1.4 in .\AppData\Local\Programs\Python\P
ython313\Lib\site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython>=7.23.1->i
pykernel!=6.30.0,>=6.5.0->jupyterlab<5,>=4->jupyterlab_execute_time) (0.9.2)
Requirement already satisfied: parso<0.9.0,>=0.8.7 in .\AppData\Local\Programs\Pyt
hon\Python313\Lib\site-packages (from jedi>=0.18.2->ipython>=7.23.1->ipykernel!=6.
30.0,>=6.5.0->jupyterlab<5,>=4->jupyterlab_execute_time) (0.8.7)
Requirement already satisfied: MarkupSafe>=2.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from jinja2>=3.0.3->jupyterlab<5,>=4->jupyterlab_exe
cute_time) (3.0.4)
Requirement already satisfied: attrs>=22.2.0 in .\AppData\Local\Programs\Python\Py
thon313\Lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->
jupyterlab<5,>=4->jupyterlab_execute_time) (26.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in .\AppData\L
ocal\Programs\Python\Python313\Lib\site-packages (from jsonschema>=4.18.0->jupyter
lab-server<3,>=2.28.0->jupyterlab<5,>=4->jupyterlab_execute_time) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in .\AppData\Local\Programs\Pyt
hon\Python313\Lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.
28.0->jupyterlab<5,>=4->jupyterlab_execute_time) (0.37.0)
Requirement already satisfied: rpds-py>=0.25.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.
0->jupyterlab<5,>=4->jupyterlab_execute_time) (2026.9.1)
Requirement already satisfied: python-dateutil>=2.8.2 in .\AppData\Local\Programs
\Python\Python313\Lib\site-packages (from jupyter-client>=7.4.4->jupyter-server<3,
>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (2.9.0.post0)
Requirement already satisfied: platformdirs>=2.5 in .\AppData\Local\Programs\Pytho
n\Python313\Lib\site-packages (from jupyter-core->jupyterlab<5,>=4->jupyterlab_exe
cute_time) (4.13.5)
Requirement already satisfied: python-json-logger>=2.0.4 in .\AppData\Local\Progra
ms\Python\Python313\Lib\site-packages (from jupyter-events>=0.11.0->jupyter-server
<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (4.2.0)
Requirement already satisfied: pyyaml>=5.3 in .\AppData\Local\Programs\Python\Pyth
on313\Lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->j
upyterlab<5,>=4->jupyterlab_execute_time) (6.0.3)
Requirement already satisfied: rfc3339-validator in .\AppData\Local\Programs\Pytho
n\Python313\Lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.1
9.0->jupyterlab<5,>=4->jupyterlab_execute_time) (0.1.4)
Requirement already satisfied: rfc3986-validator>=0.1.1 in .\AppData\Local\Program
s\Python\Python313\Lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<
```

```
3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (0.1.1)
Requirement already satisfied: fqdn in .\AppData\Local\Programs\Python\Python313\L
ib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->
jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (1.6.0)
Requirement already satisfied: isoduration in .\AppData\Local\Programs\Python\Pyth
on313\Lib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=
0.11.0->jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (20.
11.0)
Requirement already satisfied: jsonpointer>1.13 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-even
ts>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time)
(3.4)
Requirement already satisfied: rfc3987-syntax>=1.1.0 in .\AppData\Local\Programs\P
ython\Python313\Lib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter
-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_t
ime) (1.1.0)
Requirement already satisfied: uri-template in .\AppData\Local\Programs\Python\Pyt
hon313\Lib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=
0.11.0->jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (1.
3.0)
Requirement already satisfied: webcolors>=24.6.0 in .\AppData\Local\Programs\Pytho
n\Python313\Lib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-eve
nts>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time)
(25.10.0)
Requirement already satisfied: beautifulsoup4 in .\AppData\Local\Programs\Python\P
ython313\Lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.19.0->jupy
terlab<5,>=4->jupyterlab_execute_time) (4.15.0)
Requirement already satisfied: bleach!=5.0.0 in .\AppData\Local\Programs\Python\Py
thon313\Lib\site-packages (from bleach[css]!=5.0.0->nbconvert>=6.4.4->jupyter-serv
er<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (6.4.0)
Requirement already satisfied: defusedxml in .\AppData\Local\Programs\Python\Pytho
n313\Lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.19.0->jupyterl
ab<5,>=4->jupyterlab_execute_time) (0.7.1)
Requirement already satisfied: jupyterlab-pygments in .\AppData\Local\Programs\Pyt
hon\Python313\Lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.19.0-
>jupyterlab<5,>=4->jupyterlab_execute_time) (0.3.0)
Requirement already satisfied: mistune<4,>=2.0.3 in .\AppData\Local\Programs\Pytho
n\Python313\Lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.19.0->j
upyterlab<5,>=4->jupyterlab_execute_time) (3.4.0)
Requirement already satisfied: nbclient>=0.5.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.19.0->ju
pyterlab<5,>=4->jupyterlab_execute_time) (0.11.0)
Requirement already satisfied: pandocfilters>=1.4.1 in .\AppData\Local\Programs\Py
thon\Python313\Lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.19.0
->jupyterlab<5,>=4->jupyterlab_execute_time) (1.5.1)
Requirement already satisfied: webencodings in .\AppData\Local\Programs\Python\Pyt
hon313\Lib\site-packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert>=6.4.4
->jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (0.6.1)
Requirement already satisfied: tinycss2>=1.1.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from bleach[css]!=5.0.0->nbconvert>=6.4.4->jupyter-s
erver<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (1.5.1)
Requirement already satisfied: fastjsonschema>=2.15 in .\AppData\Local\Programs\Py
thon\Python313\Lib\site-packages (from nbformat>=5.3.0->jupyter-server<3,>=2.19.0-
>jupyterlab<5,>=4->jupyterlab_execute_time) (2.22.2)
Requirement already satisfied: six>=1.5 in .\AppData\Local\Programs\Python\Python3
13\Lib\site-packages (from python-dateutil>=2.8.2->jupyter-client>=7.4.4->jupyter-
server<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in .\AppData\Local\Program
s\Python\Python313\Lib\site-packages (from requests>=2.31->jupyterlab-server<3,>=
2.28.0->jupyterlab<5,>=4->jupyterlab_execute_time) (3.5.2)
Requirement already satisfied: urllib3<3,>=1.26 in .\AppData\Local\Programs\Python
```

```
\Python313\Lib\site-packages (from requests>=2.31->jupyterlab-server<3,>=2.28.0->j
upyterlab<5,>=4->jupyterlab_execute_time) (2.8.0)
Requirement already satisfied: lark>=1.2.2 in .\AppData\Local\Programs\Python\Pyth
on313\Lib\site-packages (from rfc3987-syntax>=1.1.0->jsonschema[format-nongpl]>=4.
18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyter
lab_execute_time) (1.3.1)
Requirement already satisfied: executing>=1.2.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from stack_data>=0.6.0->ipython>=7.23.1->ipykernel!=
6.30.0,>=6.5.0->jupyterlab<5,>=4->jupyterlab_execute_time) (2.3.0)
Requirement already satisfied: asttokens>=2.1.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from stack_data>=0.6.0->ipython>=7.23.1->ipykernel!=
6.30.0,>=6.5.0->jupyterlab<5,>=4->jupyterlab_execute_time) (3.0.2)
Requirement already satisfied: pure-eval in .\AppData\Local\Programs\Python\Python
313\Lib\site-packages (from stack_data>=0.6.0->ipython>=7.23.1->ipykernel!=6.30.0,
>=6.5.0->jupyterlab<5,>=4->jupyterlab_execute_time) (0.2.4)
Requirement already satisfied: cffi>=1.0.1 in .\AppData\Local\Programs\Python\Pyth
on313\Lib\site-packages (from argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-ser
ver<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (2.1.1)
Requirement already satisfied: pycparser in .\AppData\Local\Programs\Python\Python
313\Lib\site-packages (from cffi>=1.0.1->argon2-cffi-bindings->argon2-cffi>=21.1->
jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (3.11)
Requirement already satisfied: soupsieve>=1.6.1 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from beautifulsoup4->nbconvert>=6.4.4->jupyter-serve
r<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_execute_time) (3.0.3)
Requirement already satisfied: arrow>=0.15.0 in .\AppData\Local\Programs\Python\Py
thon313\Lib\site-packages (from isoduration->jsonschema[format-nongpl]>=4.18.0->ju
pyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyterlab_exec
ute_time) (1.4.0)
Requirement already satisfied: tzdata in .\AppData\Local\Programs\Python\Python313
\Lib\site-packages (from arrow>=0.15.0->isoduration->jsonschema[format-nongpl]>=4.
18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab<5,>=4->jupyter
lab_execute_time) (2026.5)
Note: you may need to restart the kernel to use updated packages.
```

In [42]:
```
pip install matplotlib
```

```
Collecting matplotlib
  Downloading matplotlib-3.11.2-cp313-cp313-win_amd64.whl.metadata (80 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.4.0-cp313-cp313-win_amd64.whl.metadata (3.8 kB)
Collecting cycler>=0.10 (from matplotlib)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib)
  Downloading fonttools-4.67.0-cp313-cp313-win_amd64.whl.metadata (139 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.5.1-cp313-cp313-win_amd64.whl.metadata (5.2 kB)
Collecting numpy>=1.25 (from matplotlib)
  Downloading numpy-2.5.4-cp313-cp313-win_amd64.whl.metadata (6.6 kB)
Requirement already satisfied: packaging>=20.0 in .\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from matplotlib) (26.3)
Collecting pillow>=9 (from matplotlib)
  Downloading pillow-13.0.0-cp313-cp313-win_amd64.whl.metadata (9.3 kB)
Collecting pyparsing>=3 (from matplotlib)
  Downloading pyparsing-3.3.3-py3-none-any.whl.metadata (5.9 kB)
Requirement already satisfied: python-dateutil>=2.7 in .\AppData\Local\Programs\Py
thon\Python313\Lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in .\AppData\Local\Programs\Python\Python3
13\Lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Downloading matplotlib-3.11.2-cp313-cp313-win_amd64.whl (9.3 MB)
   ---------------------------------------- 0.0/9.3 MB ? eta -:--:--
   ------------------ --------------------- 4.5/9.3 MB 26.4 MB/s eta 0:00:01
   ---------------------------------------- 9.3/9.3 MB 32.3 MB/s  0:00:00
Downloading contourpy-1.4.0-cp313-cp313-win_amd64.whl (234 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.67.0-cp313-cp313-win_amd64.whl (2.5 MB)
   ---------------------------------------- 0.0/2.5 MB ? eta -:--:--
   ---------------------------------------- 2.5/2.5 MB 49.2 MB/s  0:00:00
Downloading kiwisolver-1.5.1-cp313-cp313-win_amd64.whl (70 kB)
Downloading numpy-2.5.4-cp313-cp313-win_amd64.whl (12.6 MB)
   ---------------------------------------- 0.0/12.6 MB ? eta -:--:--
   ----------------------------- ---------- 9.2/12.6 MB 45.6 MB/s eta 0:00:01
   ---------------------------------------- 12.6/12.6 MB 43.7 MB/s  0:00:00
Downloading pillow-13.0.0-cp313-cp313-win_amd64.whl (7.3 MB)
   ---------------------------------------- 0.0/7.3 MB ? eta -:--:--
   ---------------------------------------- 7.3/7.3 MB 40.7 MB/s  0:00:00
Downloading pyparsing-3.3.3-py3-none-any.whl (126 kB)
Installing collected packages: pyparsing, pillow, numpy, kiwisolver, fonttools, cy
cler, contourpy, matplotlib

   ---------------------------------------- 0/8 [pyparsing]
   ----- ---------------------------------- 1/8 [pillow]
   ----- ---------------------------------- 1/8 [pillow]
   ----- ---------------------------------- 1/8 [pillow]
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   ---------- ----------------------------- 2/8 [numpy]
```

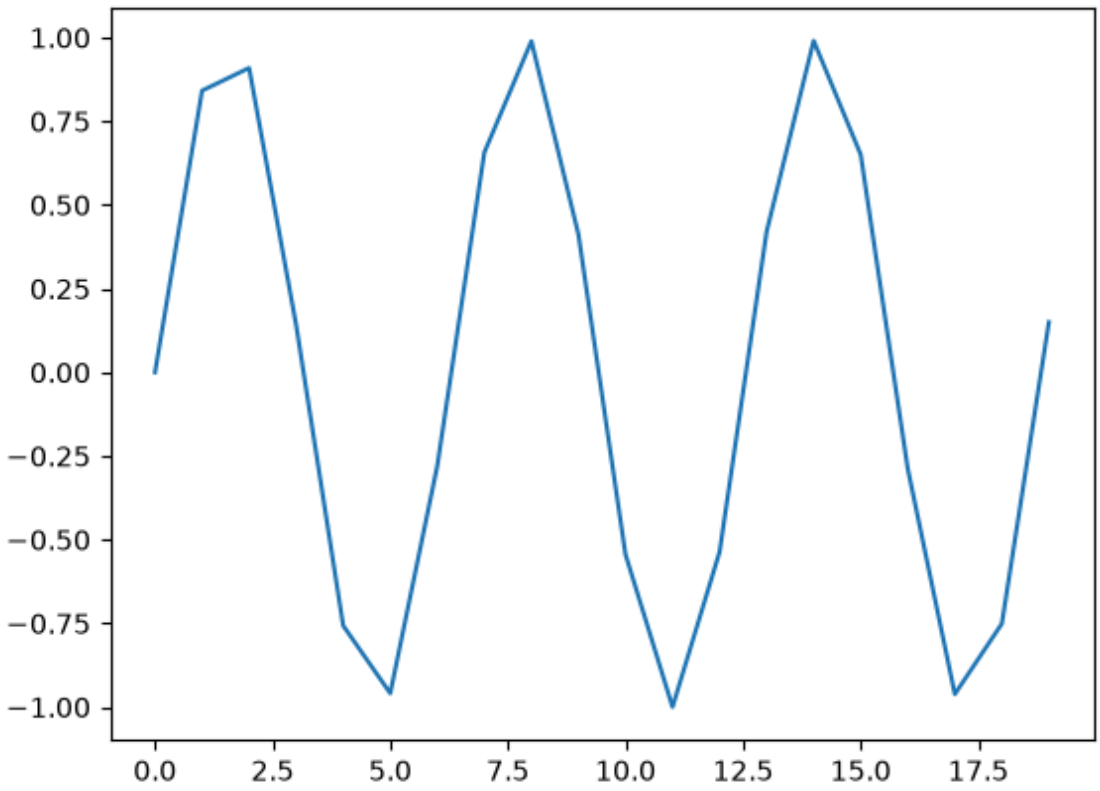
```
---------- ---------------------------- 2/8 [numpy]
---------- ---------------------------- 2/8 [numpy]
---------- ---------------------------- 2/8 [numpy]
```

```
     ---------------------------------- ---- 7/8 [matplotlib]
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     ---------------------------------- ---- 7/8 [matplotlib]
     ---------------------------------- ---- 7/8 [matplotlib]
     ---------------------------------- ---- 8/8 [matplotlib]

Successfully installed contourpy-1.4.0 cycler-0.12.1 fonttools-4.67.0 kiwisolver-
1.5.1 matplotlib-3.11.2 numpy-2.5.4 pillow-13.0.0 pyparsing-3.3.3
Note: you may need to restart the kernel to use updated packages.
```

In [43]:
```
import matplotlib.pyplot as plt
import numpy as np
```

In [44]:
```
plt.plot (np.sin(range(20)));
```




In [ ]:
```

```